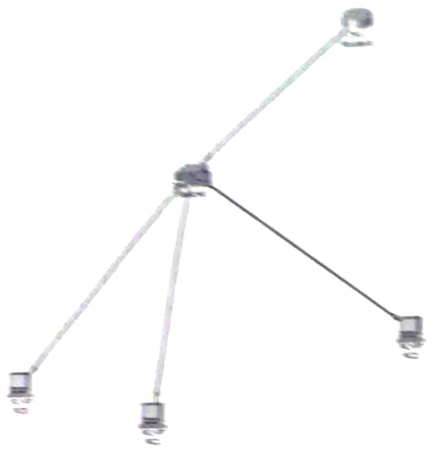
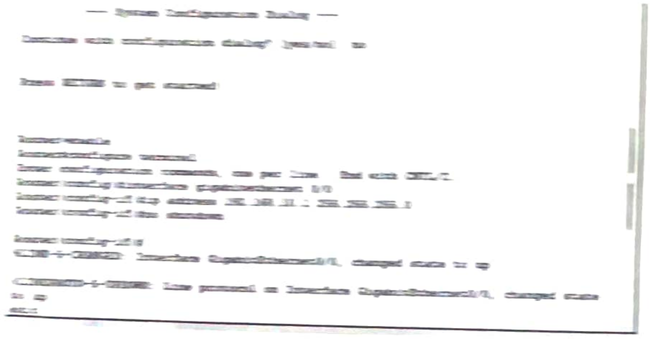




[illegible]

Implementation of Subnetting in CISCO PACKET TRACER

AIM:

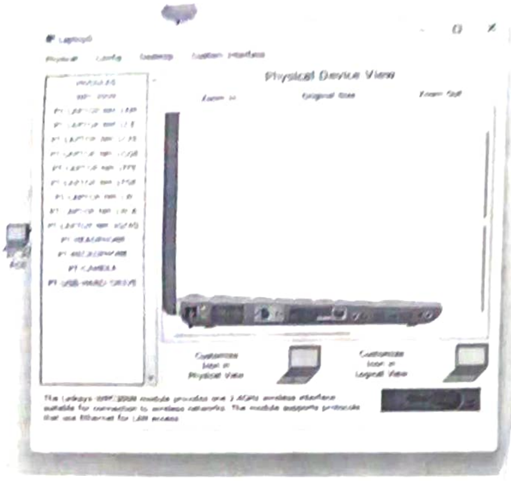
To implement subnetting by dividing a large network into multiple smaller subnetwork according to required number of host.

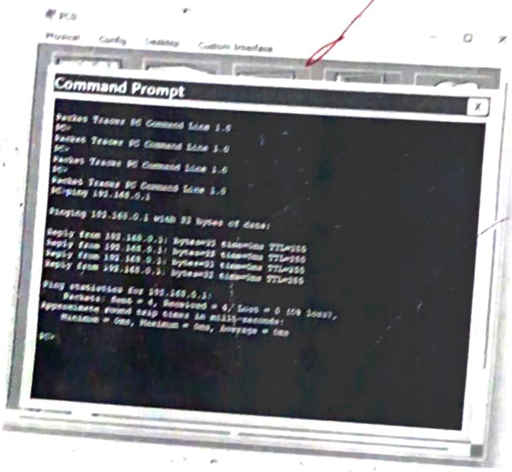
Steps:

1) Open cisco packet Tracer and create a new project
2) Design the network topology by placing required network device
3) Select a suitable network address
4) Calculate number of subnet.
5) Determine no. of hosts needed
6) Calculate subnet mask
7) Divide the network into multiple
8) Assign a separate network address
9) Verify the subnet calculation
10) Save the network topology.

Result:

Thus multiple subnets were successfully created according to the required number of hosts, and the network address was efficiently divided.





```
Packet Tracer PC Command Line 1.0
PC>
Packet Tracer PC Command Line 1.0
PC>
Packet Tracer PC Command Line 1.0
PC>
Packet Tracer PC Command Line 1.0
PC>ping 192.168.0.1

Pinging 192.168.0.1 with 32 bytes of data:

Reply from 192.168.0.1: bytes=32 time=0ms TTL=255
Reply from 192.168.0.1: bytes=32 time=0ms TTL=255
Reply from 192.168.0.1: bytes=32 time=0ms TTL=255
Reply from 192.168.0.1: bytes=32 time=0ms TTL=255

Ping statistics for 192.168.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

PC>
```

EXP 9b
7/8

ASSIGN STATIC IP ADDRESS ALL SUBNET AND CONNECT

Aim :

To assign Static IP addresses to all device in different subnet.

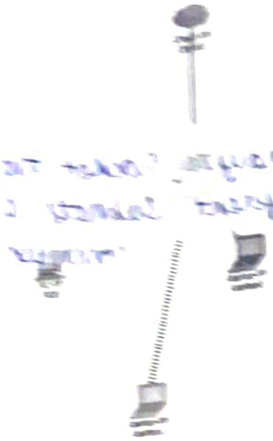
Steps :

1) Place a router in network topology
2) Connect the router to switch
3) Configure the router interfaces
4) Assign an IP address and subnet.
5) Configure the Subnet mask for each.
6) Assign the appropriate default gateway.
7) Configure the subnet mask for each.
8) Assign the appropriate gateway.
9) Verify the IP configuration of all devices
10) Save the router configuration.
11) Check the network topology.

8

Result :

Thus static IP address were successfully assigned to all devices and communication between different subnet was established.



Simulate Packet Transmission Across the Subnet

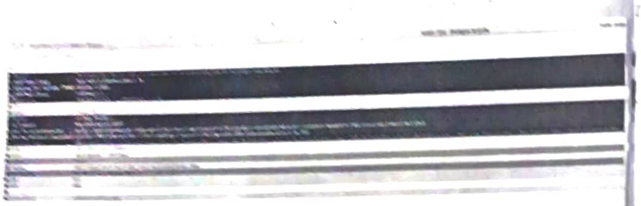
[illegible] transmission [illegible] when a [illegible] is used for [illegible]ication.

Steps:

1) Open Simulation Mode In CISCO Packet Tracer.
2) Select a source device from one Subnet.
3) Select a destination device from another subnet.
4) Generate a packet using PDU
5) Observe the movement of the packet through network.
6) Analyze how router [illegible]ives [illegible] the forwarding process
[illegible] [illegible]ify the packet transmission [illegible] using [illegible] command [illegible] connectivity
[illegible] [illegible] the Successful delivery of packet.

Result:

Thus packets were succesfully transmitted between different subnets through the router.



Ex 9c
7/8 (iii)

When Subnets are Not Connected through a Router

Aim:

To analyze packet transmission between different subnets when no router is available.

Steps:

1) Remove the router from topology
2) Open Simulation Mode
3) Select devices belonging to different subnets.
4) Generate packets using the simple PDU tool.
5) observe the packet transmission
6) perform a ping test.
7) Analyze the packet flow
8) observe the simulation events.
9) Identify the point at which packet transmission fails.
10) Compare the result with the network that uses a router.
11) Record the observation

Result:

Thus communication between devices located in different subnets failed when a router was not used.